

[Note: Every Hermitian operator on fin. dim inn. prod space is diagonalizable by its eigenvectors.
In particular, in $\mathbb{R}$, it is necessary to satisfy the condition Normal

[Spectral Theorem

Let $T: V \rightarrow V$ be a Hermitian or Normal (in $\mathbb{R}$).

Let $\lambda_1, \dots, \lambda_k$ be distinct eigenvalues for T , and $W_i = \ker(T - \lambda_i I)$, $1 \leq i \leq k$.

Let T_i be an orthogonal projection onto W_i . Then,

$$1). V = W_1 \oplus \dots \oplus W_k$$

$$2). W_i^\perp = \bigoplus_{j \neq i} W_j$$

$$3). T_i \circ T_j = \delta_{ij} T_i$$

$$4). I = T_1 + \dots + T_k$$

$$5). T = \lambda_1 T_1 + \dots + \lambda_k T_k.$$

[Corollary. In $\mathbb{C}$,

T is normal if and only if $T^* = g(T)$ for some polynomial g .

Proof. Let $g(x) = \sum_{i=1}^k \left[\frac{\prod_{j \neq i} (x - \lambda_j)}{\prod_{j \neq i} (\lambda_i - \lambda_j)} \right] \cdot \overline{\lambda_i}$.

Then $g(\lambda_i) = \overline{\lambda_i}$, so $g(T) = \sum \overline{\lambda_i} T_i = T^*$.

[Corollary. In $\mathbb{C}$,

T is unitary if and only if T is normal and every eigenvalue λ of T has $|\lambda| = 1$.

[Corollary. In $\mathbb{C}$

T is unitary if and only if T is normal and every eigenvalue is real.

Thm. Singular Value decomposition.

Let V and W be fin. dim inner product spaces, and $T: V \rightarrow W$ be a rank r linear operator. Then, there exist unique positive scalar $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$ and orthonormal basis $\{v_1, v_2, \dots, v_n\}$ and $\{u_1, \dots, u_m\}$ for V and W .

$$T(v_i) = \begin{cases} \sigma_i \cdot u_i & (1 \leq i \leq r) \\ 0 & (i > r) \end{cases}$$

Proof. T^*T is semi-positive, because $\langle T^*T\alpha, \alpha \rangle = \|T\alpha\|^2$. So all eigenvalues are positive. And, rank $T^*T = \text{rank } T$, because given $\alpha \in V$, $T\alpha = 0 \Rightarrow T^*T\alpha = 0$, so $\ker T \subseteq \ker T^*T$. Conversely, if $T^*T(\alpha) = 0$, then $0 = \langle 0, \alpha \rangle = \langle T^*T(\alpha), \alpha \rangle = \|T(\alpha)\|^2 = 0$, so $\ker T^*T \subseteq \ker T$. By the dimension Theorem, the rank are same.]

Now, since T^*T is Hermitian, there is an orthonormal eigenvectors basis for V , $\{v_1, \dots, v_n\}$, each v_i 's corresponding to positive eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_r > 0$. For each $1 \leq i \leq r$, define

$$\sigma_i = \sqrt{\lambda_i} \quad \text{and} \quad u_i = \frac{1}{\sigma_i} \cdot T(v_i).$$

Then, for each $1 \leq i, j \leq r$,

$$\begin{aligned} \langle u_i, u_j \rangle &= \left\langle \frac{1}{\sigma_i} \cdot T(v_i), \frac{1}{\sigma_j} \cdot T(v_j) \right\rangle \\ &= \frac{1}{\sigma_i \sigma_j} \cdot \langle \underbrace{T^*T(v_i)}_{=\lambda_i v_i}, v_j \rangle \\ &= \frac{\sigma_i^2}{\sigma_i \sigma_j} \langle v_i, v_j \rangle = \delta_{ij}. \end{aligned}$$

Now, extend $\{u_1, \dots, u_r\}$ to $\{u_1, \dots, u_r, \dots, u_m\}$ using Gram-Schmidt, proved.]

Corollary. Any $A \in M_{m \times n}(\mathbb{C})$, there exists unitary $U \in M_{m \times m}$, $V \in M_{n \times n}$ s.t.

$$A = U \Sigma V^* \quad \text{where} \quad \Sigma = \begin{pmatrix} \sigma_1 & & 0 \\ & \ddots & \\ 0 & & \sigma_r & \\ & & & 0 \end{pmatrix}, \quad r = \text{rank } A.$$

Proof. Let $\mathcal{B} = \{v_1, \dots, v_n\}$ be an orthonormal basis for $\mathbb{C}^n$ consisting by eigenvectors of A^*A , and for this eigenvalues λ_i , put $u_i = (\sqrt{\lambda_i})^{-1} \cdot A v_i$, and $\mathcal{B}' = \{u_1, \dots, u_r, \dots, u_m\}$, the $u_{r+1}, \dots, u_m$ are obtained by Gram-Schmidt process. Then, for standard basis α, α' ,

$$[L_A]_{\mathcal{B}}^{\mathcal{B}'} = [I_m]_{\alpha'}^{\mathcal{B}'}, [L_A]_{\alpha}^{\alpha'} [I_n]_{\mathcal{B}}^{\alpha} = \Sigma, \quad \text{so Proved.}$$

Example.

Let $V = P_2(\mathbb{R})$ and $W = P_1(\mathbb{R})$, with inner product

$$\langle f, g \rangle = \int_{-1}^1 f(t) \cdot g(t) dt.$$

Consider the differentiation $D: V \rightarrow W: f \mapsto Df$.

Using Gram-Smidt Process, find an orthonormal basis for V as:

$$d_1 = 1$$

$$d_2 = x - \frac{\langle x, 1 \rangle}{\|1\|^2} \cdot 1 = x$$

$$d_3 = x^2 - \frac{\langle x^2, 1 \rangle}{\|1\|^2} \cdot 1 - \frac{\langle x^2, x \rangle}{\|x\|^2} x = x^2 - \frac{1}{3} - 36 \cdot x \cdot 0 = x^2 - \frac{1}{3}.$$

Now, by Normalizing, we obtain

$$\beta = \left\{ \frac{1}{\sqrt{2}}, \sqrt{\frac{3}{2}} x, \sqrt{\frac{5}{8}} (3x^2 - 1) \right\}, \quad \beta' = \left\{ \frac{1}{\sqrt{2}}, \sqrt{\frac{3}{2}} x \right\} \text{ are basis for } V \text{ and } W.$$

$$\text{So, } [D]_{\beta'}^{\beta} = \begin{pmatrix} 0 & \sqrt{3} & 0 \\ 0 & 0 & \sqrt{15} \end{pmatrix}$$

$$\text{Now, } [D^* D]_{\beta} = \begin{pmatrix} 0 & 0 \\ \sqrt{3} & 0 \\ 0 & \sqrt{15} \end{pmatrix} \begin{pmatrix} 0 & \sqrt{3} & 0 \\ 0 & 0 & \sqrt{15} \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 15 \end{pmatrix}$$

So, the singular values are $\sqrt{15} \geq \sqrt{3} \geq 0 \dots$

Let $A = \begin{pmatrix} 1 & 1 & -1 \\ 1 & 1 & -1 \end{pmatrix} \in M_{2 \times 3}(\mathbb{R})$ be given.

Then, $A^* A = A^T A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ -1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 & -1 \\ 1 & 1 & -1 \end{pmatrix} = \begin{pmatrix} 2 & 2 & -2 \\ 2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}.$

$$\begin{pmatrix} 2-t & 2 & -2 \\ 2 & 2-t & -2 \\ -2 & -2 & 2-t \end{pmatrix} = \begin{pmatrix} 2-t & 2 & -2 \\ 2 & 2-t & -2 \\ -2 & -2 & 2-t \end{pmatrix} = (2-t) \cdot ((2-t)^2 - 4) - 2 \cdot (4 - 2t - 4) - 2 \cdot (-4 + 4 - 2t)$$

$$= (2-t) \cdot (-4t + t^2) - 2 \cdot (-2t) - (-2t) \\ = -8t + 2t^2 + 4t^2 - t^3 + 8t \\ = -t^3 + 6t^2 = -t^2(t-6).$$

$$\begin{pmatrix} 2 & 2 & -2 \\ -2 & -2 & 2 \\ 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & -1 \\ -1 & -1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \quad \begin{pmatrix} -4 & 2 & -2 \\ 2 & -4 & -2 \\ -2 & -2 & -4 \end{pmatrix}$$

$$\ker A^T A = \text{span} \left\{ \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \right\},$$

$$\ker (A^T A - 6I) = \text{span} \left\{ \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} \right\}.$$

Now, let $V = \left(\frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}, \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} \right)$ and $\Sigma = \begin{pmatrix} \sqrt{6} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$

And, $u_1 = \frac{1}{\sqrt{6}} \cdot \frac{1}{\sqrt{3}} \begin{pmatrix} 3 \\ 3 \\ 3 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, u_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, (\sigma_2 = \dots = 0, \text{ so } u_2 \text{ is chosen by } \langle u_1, u_2 \rangle = 0)$

Now, $A = U \Sigma V^*$, where $U = (u_1, u_2)$.

Exercise

Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be an invertible linear operator, and $S = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$. Since T is invertible, T has rank 2, therefore it has two singular values:

$\sigma_1 \geq \sigma_2 > 0$ and orthonormal basis $\beta = \{v_1, v_2\}$ and $\beta' = \{u_1, u_2\}$ s.t.

$$T(v_1) = \sigma_1 u_1 \quad \text{and} \quad T(v_2) = \sigma_2 u_2.$$

✓ Polar decomposition.

For any square $A \in M_{n \times n}(\mathbb{C})$, there exists unitary U and semi-positive P such that
$$A = UP.$$

In particular, if A is invertible, this decomposition is unique.

Proof. $A = U \Sigma V^* = \underbrace{U V^*}_W \underbrace{V \Sigma V^*}_P.$